

ST. JOSEPH COLLEGE OF ENGINEERING AND TECHNOLOGY
NTA LEVEL 5 - ORDINARY DIPLOMA IN ENGINEERING
CONTINUOUS ASSESSMENT TEST - I - JANUARY 2026
EDT 05102 - ENGINEERING MATHEMATICS

SEMESTER: I

YEAR: I

DURATION: 1 Hr 30 Min.

MAX. MARKS: 50

Auth: 4LAZIE

PART - A (Answer all the Questions) (7 x 2 = 14)

1. State and explain the sine rule.
2. Define similar figures and give an example.
3. State Heron's formula.
4. Distinguish between similarity and congruence of triangles.
5. Define differentiation.
6. Find dy/dx , if $y = 4ax$.
7. Evaluate $\int x^3 dx$.

PART - B (Answer any three Questions) (3 x 4 = 12)

8. In a triangle sides $a=12$ cm, $b=19$ cm and $\angle A = 50^\circ$. Find $\angle B$.
9. Find the projection of side b on the base side, if $b=10$ cm and $\angle A = 0^\circ$.
10. Find the area of the triangle with sides $a=15$ cm, $b=20$ cm and $c=8$ cm.
11. Find dy/dx , if $y = \tan^{-1}\sqrt{x}$.
12. Evaluate $\int (x^2 - 2x + 3) dx$.
13. If $y = 2x^5 - 4x^2 + 3$, Find d^2y/dx^2 .

PART - C (Answer any two Questions) (2 x 12 = 24)

14. Explain the projection formula with proof.
15. The displacement of a particle is $s(t) = 5t^3 - 2t^2 + 4t$ (in metres).
Find (i) Velocity at time $t=2$ second
(ii) Acceleration at time $t=2$ second.
16. Evaluate $\int x^2 \sin^2 x dx$.
17. State and explain the AA, SAS, and SSS similarity theorems of triangles with suitable diagrams.